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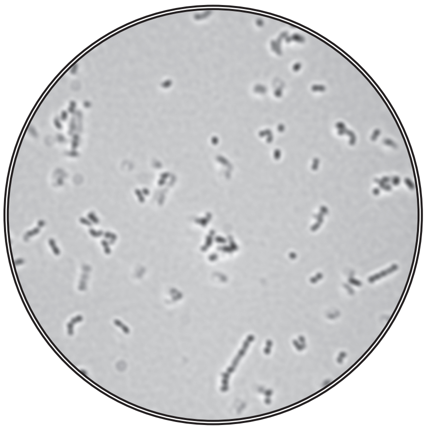
3. The Breed method is a fast and simple way of counting the number of bacteria in a suspension.

A known volume of the bacterial suspension is spread uniformly over a glass slide covering a specific area. The bacteria are then heat fixed and stained.

The photograph below shows the results of this method as seen through a light microscope.



(a) The average number of cells in the field of view was found to be 156 and the radius of the field of view was 0.09 mm.

Calculate the number of cells in 100 mm^2.

The area of microscopic field of view = πr^2

where π is 3.14 and r is the radius of the field of view. [3]

Number of cells = ...

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(b) A scientist used this method to find the number of bacteria in a sample taken from the back of a patient's throat. There were too many bacteria to be able to count down the microscope.

(i) Describe a method that could be used to produce a range of dilutions from the original sample to a 10^{-5} dilution. [3]

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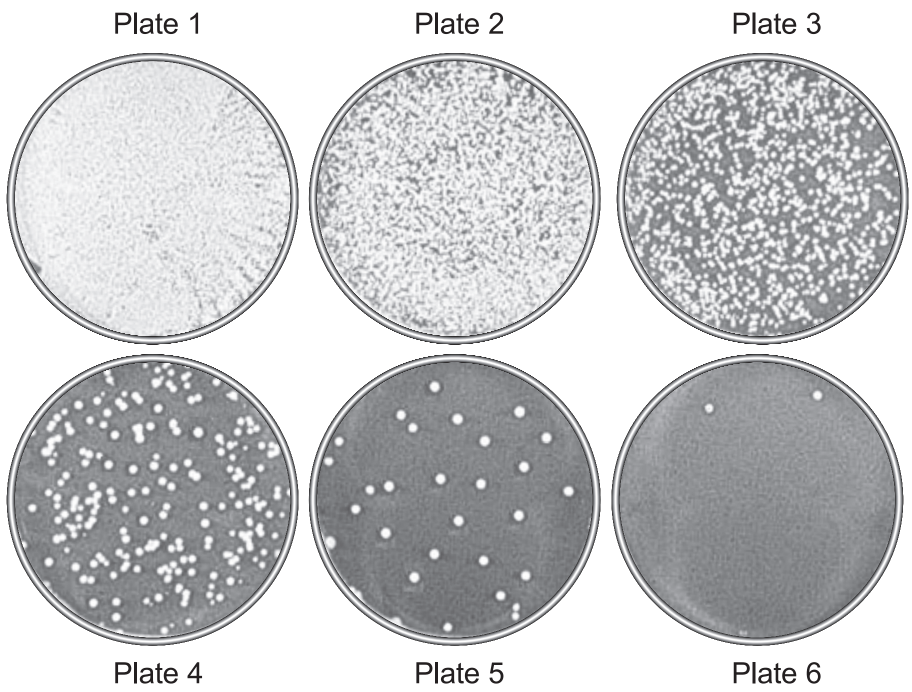
The scientist obtained the following results for a range of dilutions.



(ii) State which number plate should be used and give reasons why the other plates should not be used. [3]

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The scientist was looking for the pathogenic bacterium *Staphylococcus aureus*.

(iii) Give the reason why the bacteria removed from the patient were incubated at 37 °C not 25 °C. [1]

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(iv) A sample of the *Staphylococcus* was stained purple by the Gram stain technique. Describe what the purple staining indicates about the structure of the bacterial cell wall. [2]

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